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Finite difference thermal model of a latent heat storage system coupled with a photovoltaic device: Description and experimental validation

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ABSTRACT

The use of photovoltaic (PV) systems has been showing a significant growth trend but for a more effective development of this technology it is essential to have higher energy conversion performances. Producers of PV often declare an higher efficiency respect to real conditions and this deviation is mainly due to the difference between nominal and real temperature conditions of the PV. To improve the solar cell energy conversion efficiency many authors have proposed a methodology to keep lower the temperature of a PV system: a modified PV system built with a normal PV panel coupled with a Phase Change Material (PCM) heat storage device. In this paper is described a thermal model analysis of the PV–PCM system based on a theoretical study using finite difference approach. The authors developed an algorithm based on an explicit finite difference formulation of energy balance of the PV–PCM system. To this aim, a forward difference at time t and a first-order central difference for the space derivative at position x was used. Two sets of recursive equations were developed for two types of spatial domains: a boundary domain and an internal domain .The reliability of the developed model is tested by a comparison with data coming from a test facility. Results of this experience confirm the performed numerical simulations and show that the proposed model is valid and can be used to determine the thermal behaviour of a solar cell coupled with a PCM heat storage device.

1. Introduction

The intense exploitation of fossil fuels has caused an increase in the concentration of carbon dioxide from 280 ppm to 370 ppm and a consequent estimated global warming of 0.4–0.8 °C. The global effort to fight the effects of global climate change has been oriented at reducing energy consumptions through new technologies [1] and ensuring a more efficient exploitation of RES (Renewable Energy Sources) [2].

Among RES, solar energy is the most important and available natural resource. During the last decade there was a widespread use of photovoltaic (PV) systems not only for decentralized production in advanced countries but also in developing countries, where the most likely alternative to produce electricity is related with the use of poor fuels (coal and lignite, peat, etc. which are very polluting) and biomass.

A key element of a wider dissemination of PV systems is represented by a high power conversion efficiency. Concerning this point, the energy produced by a PV cell depends, apart materials, also on other two important parameters: the amount of the incident radiation and the temperature of the PV cell.

The performances of a PV panel in fact are defined by manufactures according to the "peak power", which identifies the maximum electric power supplied by the PV panel when it receives an insolation of 1 kW/m^2 and the cell temperature is maintained at 25 °C. These parameters are only observed under reference conditions, because solar radiation has a variable intensity and the panel is subjected to significant temperature changes, with temperature values much higher than 25 °C. In real conditions performances of a PV panel are different from those declared under the nominal conditions and the conversion efficiency decreases when temperature of the cell increases [3].

For given values of insolation, cell temperature and electric load the operating point can be identified by drawing lines of the different loads on the P–V characteristic (Fig. 1); the maximum

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power points are indicated by red circles (colour in the web version).

The cell temperature thus is a key parameter that affects the energy conversion efficiency of a PV panel: increasing the temperature decreases the delivered power.

The wind speed greatly influences the heat exchange between the PV panel and the external environment mitigating its temperature; however, especially in densely populated urban areas often the wind is too weak [4–6] to determine a desirable cooling of the solar cell.

Among other measures aimed to increase the energy conversion, PCMs have been receiving increased attention, due to their capacity to store large amounts of thermal energy in narrow temperature range. PCMs represent a possible solution that may reduce peak loads and thermal energy consumption in buildings due to their good insulation properties and thermal inertia effect related to the phase change phenomenon [7–9]. Indeed this property makes them ideal for passive heat storage in different building application as: in the envelope of the building, in radiant floor systems heating, in free cooling system, in the photovoltaic element and in building integrated PV [10–18].

The idea to couple the PCMs with the photovoltaic technology arises from features of these materials to absorb large amounts of heat (keeping almost constant the temperature) when the heat is not required and overheating would cause a drop in the efficiency of photovoltaic cells. The absorbed heat should be then released to the surrounding air during the night when the panel does not produce electrical power. The application of PCM coupled to a PV panel may represent an innovative technological solution to smooth daily temperature fluctuations and improve the energy efficiency of the panel. However, PCMs are generally characterized by low thermal conductivity.

To better understand how PCM ensures a thermal regulation for a PV system, it is necessary to examine in detail the heat transfer process across a multi-layered system in which one of the layers is composed by a material that changes phase during the day.

In this paper a finite difference model is described, capable to forecast the variable temperature profile of a PV-PCM system; the adopted numerical scheme is presented in detail, also discussing the equations of the model and the resolution system using the FDM approach (finite difference method).

To validate the numerical model, we performed a comparison with data derived by a real-time monitoring apparatus.

2. PV–PCM model

Considering a PV panel coupled with PCM system, the energy balance must take into account the presence of the phase-change material. Schematically, the energy exchanges in a PV–PCM system can be exemplified as shown in Fig. 1. Due to the presence of a simple geometry, it was possible to adopt a one-dimensional approach, considering only a heat flow orthogonal to the PV plane. The simplification of the thermal problem giving up a more accurate representation in 2D or 3D [19] does not lead to unacceptable errors in the evaluation of the temperature field. The hypothesis of mono-dimensionality of the heat flow is in fact justified by the ratio thickness/surface, which in the case of the PV–PCM system is close to 0.02 m^{-1}. The thermo-physical characteristics of each component are also constant with respect to the other two directions and neglecting the effects relative to the edges of the panel the overall energy balance remains practically unchanged. Furthermore, with regard to the PCM, it is confined in small vacuum plastic bags about 250 ml. This configuration, also thanks to the high viscosity of the material in liquid phase allows to completely exclude the establishment of natural neither buoyant convection.

In Fig. 2h_{rad} and h_{conv}, respectively, represent the external radiative and convective coefficients.

Let us refer to a particular geometry, assuming the system composed by:

- a tempered glass sheet with a thickness of 3.2 mm (glass layer);
- 1 mm of PET plastic panel on which are "printed" the silicon cells; the silicon cells are considered having negligible thickness (plastic layer);
- an optional layer of air interposed between the panel and the heat storage system representing a possible imperfect contact (air layer);

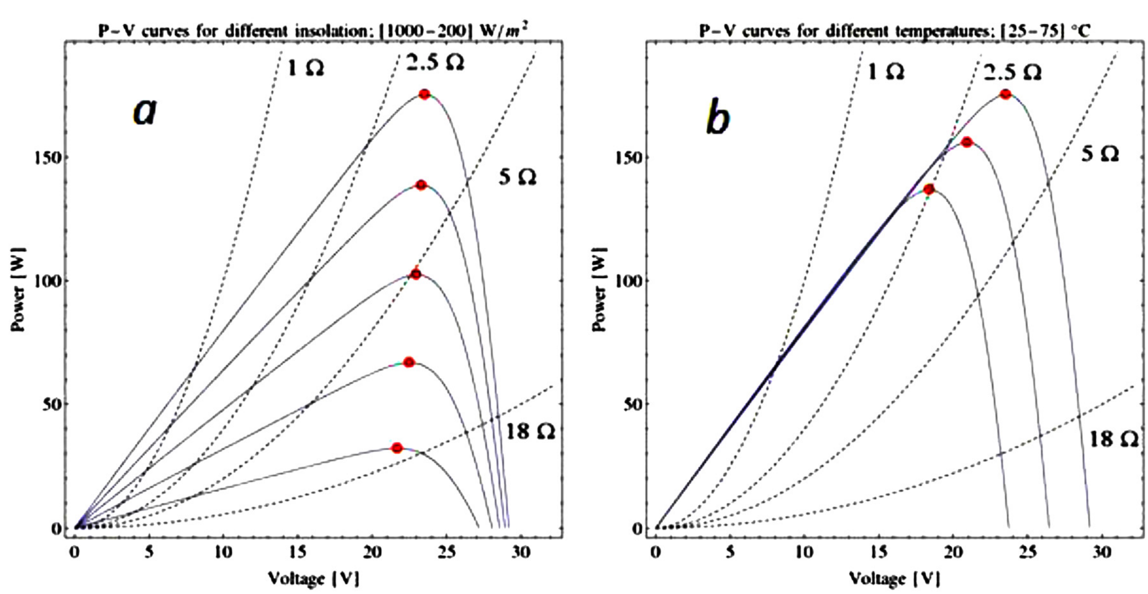

Fig. 1. Working point of the examined Kyocera PV panel at constant temperature (25 °C) and varying solar irradiance (1000–200 W/m^2) and electric load (a) and at constant irradiance (1000 W/m^2) varying temperature (25–75 °C) and electric load (b).

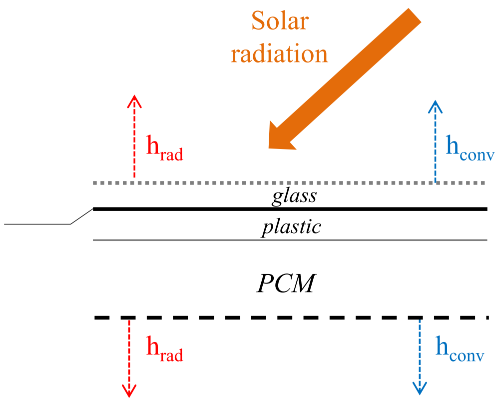

Fig. 2. Reference scheme of the PV–PCM system.

- a plastic layer that takes into account the bag that contains the PCM (bag layer);
- about 5 cm of PCM (PCM layer).

Fig. 3 represents the section of the geometry along the thickness of the PV–PCM system where it is possible to identify all the involved layers: the glass panel, the PET panel, the layer of air due to the imperfect contact of the envelope of PCM with the PV, the thickness of the envelope containing the PCM and the PCM layer itself.

The PV–PCM system is presented as a multi-layer plate invested by solar radiation and exchanging heat with the external environment by convection and radiation. Depending on the properties of the PCM and on the amount of energy captured from the panel, the PCM layer can partially or totally melt during the maximum insolation, rejecting the same amount of energy initially stored (and possibly solidifying again) during the night. The first hypothesis is that the phase change is perfectly isothermal. This hypothesis is not very far from reality because many PCMs are characterized by isothermal phase change, while some paraffin and eutectic mixtures have a very short range of temperature during transition In case of non-isothermal transition, if the c_p value is known at any temperature; the problem is reduced to a heat conduction case. The occurrence of a phase change that fixes the temperature at a given value is a more interesting case, as concerns the analysis of thermal profiles, and it implies the determination of the PCM liquid fraction.

2.1. One-dimensional thermal analysis of an isothermal phase change by using explicit finite difference approach

If we consider a one-dimensional approach with constant thermo-physical properties and without internal heat generation, assuming that the heat transfer is only due to conduction, the problem can be described by a system of equations involving two heat diffusion equations and an energy balance pertaining the PCM [20–23].

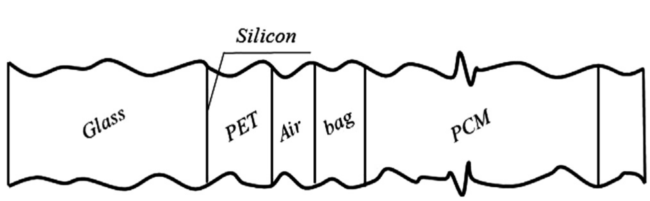

Fig. 3. PV–PCM cross-section (thicknesses not presented at real scale, for clarity of representation).

$$\left.\begin{aligned}\frac{\partial^2 T}{\partial x^2} &= \frac{1}{\alpha_s}\frac{\partial T}{\partial t}\\ \frac{\partial^2 T}{\partial x^2} &= \frac{1}{\alpha_l}\frac{\partial T}{\partial t}\\ \lambda_s\frac{\partial T}{\partial x}\Big|_{x=X(t)} - \lambda_l\frac{\partial T}{\partial x}\Big|_{x=X(t)} &= \rho_s\cdot l\frac{\mathrm{d}X(t)}{\mathrm{d}t}\end{aligned}\right\} \tag{1}$$

where the superscripts s and l refer to the solid and liquid phase, l is the specific solidification latent heat, V is the control volume and $x = X(t)$ is the time function that identifies the position of the phase change boundary.

The balance relating to system equations (1) refers to the energy balance of a homogeneous, continuous, isotropic system in one-dimensional geometry; the application of the finite difference method requires the discretization of the system both in space and time. However, in case of phase change of the medium, it is easier to use equations that are function of enthalpy variation, thus allowing to reduce the above system to a single equation.

If we assume that T_m is the temperature of the phase change and the liquid fraction is defined as:

$$f_n = \begin{cases} 1 & \text{if} \quad T_i > T_m \\ f_n(t) & \text{if} \quad T_i = T_m \\ 0 & \text{if} \quad T_i < T_m \end{cases} \tag{2}$$

the total enthalpy I can be expressed as sum of the sensible component S and the latent component Lf:

$$I = S + L\cdot f \tag{3}$$

Then, rearranging the previous expressions:

$$I = S + Lf \Rightarrow \frac{\partial I}{\partial t} = \frac{\partial S}{\partial t} + L\frac{\partial f}{\partial t} \tag{4}$$

$$\mathrm{d}S = C\,\mathrm{d}T \tag{5}$$

where C is the average heat capacity of an uncompressible substance (for small temperature variations the specific heat capacity can be assumed as constant).

In this case the previous system equation can be merged in a unique expression:

$$\frac{\partial I}{\partial t} = C\frac{\partial T}{\partial t} + L\frac{\partial f}{\partial t} \tag{6}$$

To better understand how the phase change depends on the position of node, in the following paragraphs the enthalpy balance is separately written for superficial and internal nodes for different initial conditions.

Eq. (5) permits to assess the enthalpy balance of the domain that pertains superficial and internal nodes. The finite difference approximation of the total enthalpy, can be written as

$$\frac{\partial I_n}{\partial t} \approx \frac{I_n^{p+1} - I_n^p}{\Delta t} = \frac{S_n^{p+1} - S_n^p}{\Delta t} + L\frac{f_n^{p+1} - f_n^p}{\Delta t} \tag{7}$$

where the subscript n identifies the position along the x axe of the examined nodal point and the superscript p denotes the time dependence; $p + 1$ denotes the present time while p denotes the previous time interval. To ensure the convergence of Eq. (7) the condition should be fulfilled [24]:

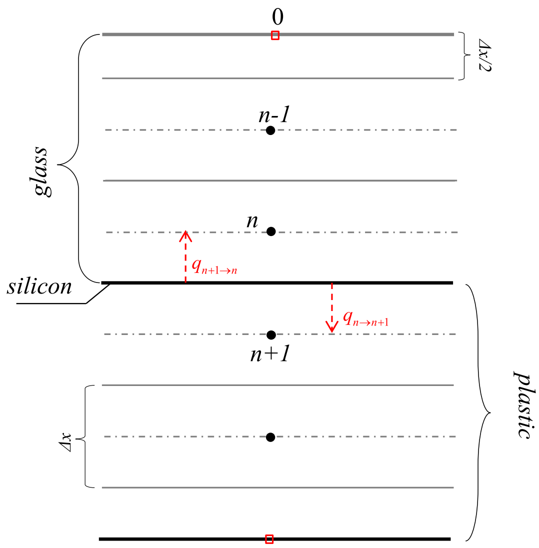

Fig. 4. Scheme of spatial discretization of PV–PCM system.

$$\frac{\Delta t}{\Delta x^2} \le \frac{\rho C}{2\lambda} \tag{8}$$

Thus, concerning the spatial discretization, the thickness of domains Δx is different for each materials, and the choice is made to fulfil the stability criterion presented in Eq. (8) (Fig. 4).

2.2. Enthalpy balance equations

The thickness of the control volume associated with a generic border node 0 is halved respect to the thickness of a generic internal node; hence, using the enthalpy notation of Eq. (6) and assuming that on the external surface there is a generic heat flux q (that takes into account the radiative exchange with surrounding environment) and a convective heat transfer process, we can state that:

$$\frac{\partial I_n}{\partial t} \approx \frac{I_n^{p+1} - I_n^p}{\Delta t} = hA\left(T_\infty^p - T_0^p\right) + 2\frac{\overline{\lambda}_{1\to 0}A}{\Delta x}\left(T_1^p - T_0^p\right) + q^p A$$

where

$$\begin{aligned}\frac{I_n^{p+1} - I_n^p}{\Delta t} &= \frac{S_0^{p+1} - S_0^p}{\Delta t} + L\frac{f_0^{p+1} - f_0^p}{\Delta t} \\ &= \rho_0 c_0 A\frac{\Delta x}{2\Delta t}\left(T_0^{p+1} - T_0^p\right) + l\rho_0 A\frac{\Delta x}{2\Delta t}\left(f_0^{p+1} - f_0^p\right)\end{aligned}$$

and then

$$\begin{aligned}&\rho_0 c_0\left(T_0^{p+1} - T_0^p\right) + l\rho_0\left(f_0^{p+1} - f_0^p\right) \\ &\quad = 2h\Delta t\frac{\left(T_\infty^p - T_0^p\right)}{\Delta x} + 4\frac{\overline{\lambda}_{1\to 0}\Delta t}{\Delta x^2}\left(T_1^p - T_0^p\right) + 2q^p\frac{\Delta t}{\Delta x}\end{aligned} \tag{9}$$

whereρ_0 is the density of the control volume pertaining the superficial node 0, [kg/m^3]c_0 is the specific heat at constant pressure pertaining the superficial node 0, [J/kg K]$\Delta x/2$ is the thickness of the superficial domain 0, [m]$\overline{\lambda}_{1\to 0} = (\lambda_0\lambda_1/\lambda_0 + \lambda_1)$ is function of thermal conductivities pertaining the superficial node 0 and the internal node 1, [W/mK]T_0^{p+1} is the temperature of the superficial node at the present time, [K]T_0^p is the temperature of the superficial node at the past time or previous time step, [K]h is the convective heat transfer coefficient[W/m^{2}K] calculated with the correlation of Furushima et al. [25]T_∞^p is the air temperature at the past time, [K]T_1^p is the temperature of the first internal node (after the superficial one) at the past time, [K]q^p is the external heat flux at the past time, [W/m^2]l is the specific latent heat, [J/kg].

A similar total enthalpy balance may be also written for a generic internal node, assuming the presence of a generic conductive heat flow coming from the previous and the next node.

Furthermore, the enthalpy/energy balance is written considering also the presence of the energy flow due to the absorption of solar irradiance by the silicon layer. Given the thinness of this layer, (about 0.3 mm) and the high thermal conductivity of this material, the silicon layer is considered as a virtual surface that separates the glass layer and the support material of the cells (plastic). For this purpose, the discretisation of the system is operated so that this separation surface (silicon) does not coincide with the position of any node; otherwise would be impossible to describe any extensive property pertaining this node and its domain. The energy deposition due to the absorption of solar irradiance is taken into account by considering two conductive heat flow: one that goes to the glass layer and the other that goes to the plastic layer. The paths among the point of origin of this energy deposition (the virtual surface of silicon), the glass (node n) and the plastic (node $n+1$) are then equal to the half of the distance between two nodes. For this reason, only in this case, the arrows do not extend between two nodes but originate from the separation surface (silicon).

Denoting by $q_{n+1\to n}$ the flow that is originated in the separation border between the nodes $n+1$ and n, we can write the total enthalpy balance in explicit discretized form; this method allows to calculate the state of a system at the successive time step "$t+1$", once calculated the state of the system at the current time "t".

+Starting from the approach followed by Ref. [26], where a simple implicit computational model for isothermal phase change was presented, the authors developed a modified algorithm based on an explicit finite difference formulation of the heat equation. To this aim, a forward difference at time t and a first-order central difference for the space derivative at position x was used. To make possible the application of the method to the PV-panel system, two sets of recursive equations have been developed for two types of spatial domains: a boundary domain characterised by a length equal to $\Delta x/2$ with a representative node placed on the surface, and an internal domain with length equal to Δx with a representative node placed in the middle. The novelty concerning the definition of a boundary domain allows us to directly consider the radiative and convective heat transfer occurring on the surface in the recursive equation. Furthermore, the presence of a generic local heat flux q is always considered to properly simulate the eventual presence of an active layer in the wall.

In the PV–PCM model such heat flux will be not null only at one border between two consecutive nodes, where is located the surface of silicon; the flux q will be obviously null elsewhere. In the hypothesis of only conductive heat flux coming from the previous and next nodes, it is possible to write that for the n-the domain:

$$\frac{I_n^{p+1}-I_n^p}{\Delta t} = \rho_n c_n A \frac{\Delta x}{\Delta t}\left(T_n^{p+1}-T_n^p\right) + l\rho_n A \frac{\Delta x}{\Delta t}\left(f_n^{p+1}-f_n^p\right) = \frac{\overline{\lambda}_{n-1\rightarrow n}A}{\Delta x}\left(T_{n-1}^p-T_n^p\right) + \frac{\overline{\lambda}_{n\leftarrow n+1}A}{\Delta x}\left(T_{n+1}^p-T_n^p\right) + q_{n-1\rightarrow n}\frac{\lambda_n}{\lambda_{n-1}+\lambda_n}A + q_{n\leftarrow n+1}\frac{\lambda_n}{\lambda_n+\lambda_{n+1}}A$$

$$\rho_n c_n\left(T_n^{p+1}-T_n^p\right) + l\rho_n\left(f_n^{p+1}-f_n^p\right) = \frac{\overline{\lambda}_{n-1\rightarrow n}\Delta t}{\Delta x^2}\left(T_{n-1}^p-T_n^p\right) + \frac{\overline{\lambda}_{n+1\rightarrow n}\Delta t}{\Delta x^2}\left(T_{n+1}^p-T_n^p\right) + q_{n-1\rightarrow n}\frac{\lambda_n\Delta t}{\Delta x(\lambda_{n-1}+\lambda_n)} + q_{n\leftarrow n+1}\frac{\lambda_n\Delta t}{\Delta x(\lambda_n+\lambda_{n+1})} \tag{10}$$

For the resolution of Eqs. (9) and (10), respectively, representing the enthalpy balance for control volumes associated with superficial and internal nodes at the time$p+1$, we should distinguish four different cases:

- I Case: no phase change;
- II Case: phase change;
- III Case: just started phase change;
- IV Case: just ending phase change.

2.2.1. I Case: fully solid or fully liquid (no phase change)

$$f_n^{p+1} = f_n^p = 0 \quad \text{or} \quad f_n^{p+1} = f_n^p = 1$$

In this case there is not any phase change and the time variation of the liquid fraction is null. Eqs. (9) and (10) can be solved by iterative method in terms of temperature:

For the superficial node:

$$T_0^{p+1} = T_0^p + 2h\Delta t\frac{\left(T_\infty^p - T_0^p\right)}{\rho_0 c_0\Delta x} + 4\frac{\overline{\lambda}_{1\rightarrow 0}\Delta t}{\rho_0 c_0\Delta x^2}\left(T_1^p - T_0^p\right) + 2q^p\frac{\Delta t}{\rho_0 c_0\Delta x} \tag{11}$$

For internal nodes:

$$T_n^{p+1} = T_n^p + \frac{\overline{\lambda}_{n-1\rightarrow n}\Delta t}{\rho_n c_n\Delta x^2}\left(T_{n-1}^p - T_n^p\right) + \frac{\overline{\lambda}_{n+1\rightarrow n}\Delta t}{\rho_n c_n\Delta x^2}\left(T_{n+1}^p - T_n^p\right) + q_{n-1\rightarrow n}\frac{\lambda_n\Delta t}{\rho_n c_n\Delta x(\lambda_{n-1}+\lambda_n)} + q_{n\leftarrow n+1}\frac{\lambda_n\Delta t}{\rho_n c_n\Delta x(\lambda_n+\lambda_{n+1})} \tag{12}$$

2.2.2. II Case: phase change

$$0 < f_n^{p+1} < 1 \quad \text{and} \quad 0 < f_n^p < 1$$

When the phase change is occurring, it is possible to make some observations. First of all, if the phase change is isothermal, during this process the temperature is locked to the value T_m as long as the liquid fraction is between 0 and 1. Obviously, during the process the variation of sensible enthalpy is null. When the phase change is starting the superficial temperature is T_m, and the unknown value is represented by the liquid fraction. In this case it is possible to state:

$$\frac{\partial S}{\partial t} = 0 \quad \text{and} \quad T_n^p = T_n^{p+1} = T_m$$

In this condition the liquid fraction is:

a) for the superficial node:

$$f_0^{p+1} = f_0^p + 2h\Delta t\frac{\left(T_\infty^p - T_m\right)}{l\rho_0\Delta x} + 4\frac{\overline{\lambda}_{1\rightarrow 0}\Delta t}{l\rho_0\Delta x^2}\left(T_1^p - T_m\right) + 2q^p\frac{\Delta t}{l\rho_0\Delta x} \tag{13}$$

b) for internal nodes:

$$f_n^{p+1} = f_n^p + \frac{\overline{\lambda}_{n-1\rightarrow n}\Delta t}{l\rho_n\Delta x^2}\left(T_{n-1}^p - T_m\right) + \frac{\overline{\lambda}_{n+1\rightarrow n}\Delta t}{l\rho_n\Delta x^2}\left(T_{n+1}^p - T_m\right) + q_{n-1\rightarrow n}\frac{\lambda_n\Delta t}{l\rho_n\Delta x(\lambda_{n-1}+\lambda_n)} + q_{n\leftarrow n+1}\frac{\lambda_n\Delta t}{l\rho_n\Delta x(\lambda_n+\lambda_{n+1})} \tag{14}$$

2.2.3. III Case: just started phase change

$$T_n^{p+1} = T_m; \quad T_n^{p+1} \neq T_n^p;$$

$$0 < f_n^{p+1} < 1 \quad \text{and} \quad f_n^p = 0 \quad \text{heat absorption} \quad T_n^{p+1} = T_m \quad \text{and} \quad T_n^p < T_m \quad \text{(just started the melting)}$$

$$0 < f_n^{p+1} < 1 \quad \text{and} \quad f_n^p = 1 \quad \text{heat release} \quad T_n^{p+1} = T_m \quad \text{and} \quad T_n^p > T_m \quad \text{(just started the solidification)}$$

An additional term has to be evaluated to consider the sensible heat that the control volume has adsorbed/released to pass from the initial temperature T_n^p to the phase change temperature $T_n^{p+1} = T_m$ during the examined time step.

c) for the superficial node:

$$f_0^{p+1} = f_0^p + 2h\Delta t\frac{\left(T_\infty^p - T_m\right)}{l\rho_0\Delta x} + 4\frac{\overline{\lambda}_{1\rightarrow 0}\Delta t}{l\rho_0\Delta x^2}\left(T_1^p - T_m\right) + 2q^p\frac{\Delta t}{l\rho_0\Delta x} - \frac{c}{l}\left(T_m - T_0^p\right) \tag{15}$$

d) for internal nodes:

$$f_n^{p+1} = f_n^p + \frac{\overline{\lambda}_{n-1\rightarrow n}\Delta t}{l\rho_n\Delta x^2}\left(T_{n-1}^p - T_m\right) + \frac{\overline{\lambda}_{n+1\rightarrow n}\Delta t}{l\rho_n\Delta x^2}\left(T_{n+1}^p - T_m\right) + q_{n-1\rightarrow n}\frac{\lambda_n\Delta t}{l\rho_n\Delta x(\lambda_{n-1}+\lambda_n)} + q_{n\leftarrow n+1}\frac{\lambda_n\Delta t}{l\rho_n\Delta x(\lambda_n+\lambda_{n+1})} - \frac{c}{l}\left(T_m - T_n^p\right) \tag{16}$$

2.2.4. IV case: just ending phase change

$$T_n^{p+1} \neq T_n^p; \quad T_n^p = T_m$$

$$0 < f_n^p < 1 \quad \text{and} \quad f_n^{p+1} = 1 \quad \text{heat absorption} \quad T_n^p = T_m; \quad T_n^{p+1} > T_m \quad \text{(just ending the melting)}$$

$$0 < f_n^p < 1 \quad \text{and} \quad f_n^{p+1} = 0 \quad \text{heat release} \quad T_n^p = T_m; \quad T_n^{p+1} < T_m \quad \text{(just ending the solidification)}$$

e) for superficial nodes:

$$T_0^{p+1} = T_0^p + 2h\Delta t\frac{(T_\infty^p - T_0^p)}{\rho_0 c_0 \Delta x} + 4\frac{\overline{\lambda}_{1\to 0}\Delta t}{\rho_0 c_0 \Delta x^2}(T_1^p - T_0^p) + 2q^p\frac{\Delta t}{\rho_0 c_0 \Delta x} - \frac{l}{c}\left(f_0^{p+1} - f_0^p\right) \quad (17)$$

f) for internal node:

$$T_n^{p+1} = T_n^p + \frac{\overline{\lambda}_{n-1\to n}\Delta t}{\rho_n c_n \Delta x^2}(T_{n-1}^p - T_n^p) + \frac{\overline{\lambda}_{n+1\to n}\Delta t}{\rho_n c_n \Delta x^2}(T_{n+1}^p - T_n^p) + q_{n-1\to n}\frac{\lambda_n \Delta t}{\rho_n c_n \Delta x(\lambda_{n-1} + \lambda_n)} + q_{n\leftarrow n+1}\frac{\lambda_n \Delta t}{\rho_n c_n \Delta x(\lambda_n + \lambda_{n+1})} - \frac{l}{c}\left(f_n^{p+1} - f_n^p\right) \quad (18)$$

When the phase change is just ending, the superficial temperature is again free to float; the last subtractive term in Eq. (17) account for the necessary latent heat to end the phase changing.

3. Description of calculation steps

The previous equations described above allow to develop an algorithm for the automatic calculation in a software. To simplify the description of the calculation procedure, the iterative equations used and solved for each time step Δt have been rewritten in the following general form:

$$AT_{n-1}^p + BT_n^{p+1} + DT_{n+1}^p = C \quad (19)$$

To obtain a stable and convergent solution in the algorithm the next steps are executed:

1. The coefficients A, B, C, D are calculated for each node taking into account the value of thermo-physical properties of the medium calculated at the previous time step: $\rho_n = \rho(T_n^p)$; $\lambda_n = \lambda(T_n^p)$; $c_n = c(T_n^p)$.
2. All nodal temperatures are calculated with an iterative approach.
3. Only for PCM nodes, the phase of each domain are evaluated; possible conditions are: fully solid, fully liquid, starting of melting, phase changing, end of melting, starting of solidification, end of solidification.
4. Once calculated the conditions, a check is performed over temperatures, to verify that the correct equations were used. If one or more checks failed, the correct equations are used and all nodal temperatures are recalculated (back to step 2).When phase change is starting, ending or occurring $T_n^{p+1} = T_m$, so the condition is:

$$\begin{cases} A_n = 0 \\ B_n = 1 \\ C_n = T_m \\ D_n = 0 \end{cases}$$

5. Once all nodal temperatures are calculated with right equations, liquid fraction for under transition PCM domains are calculated.

Figs. 5 and 6 show the condition evaluation and liquid fraction calculation flow charts for the case of rising temperature:

4. Energy production of PV system

The above algorithm for the resolution of the thermal field was applied to a photovoltaic panel coupled with a PCM thermal

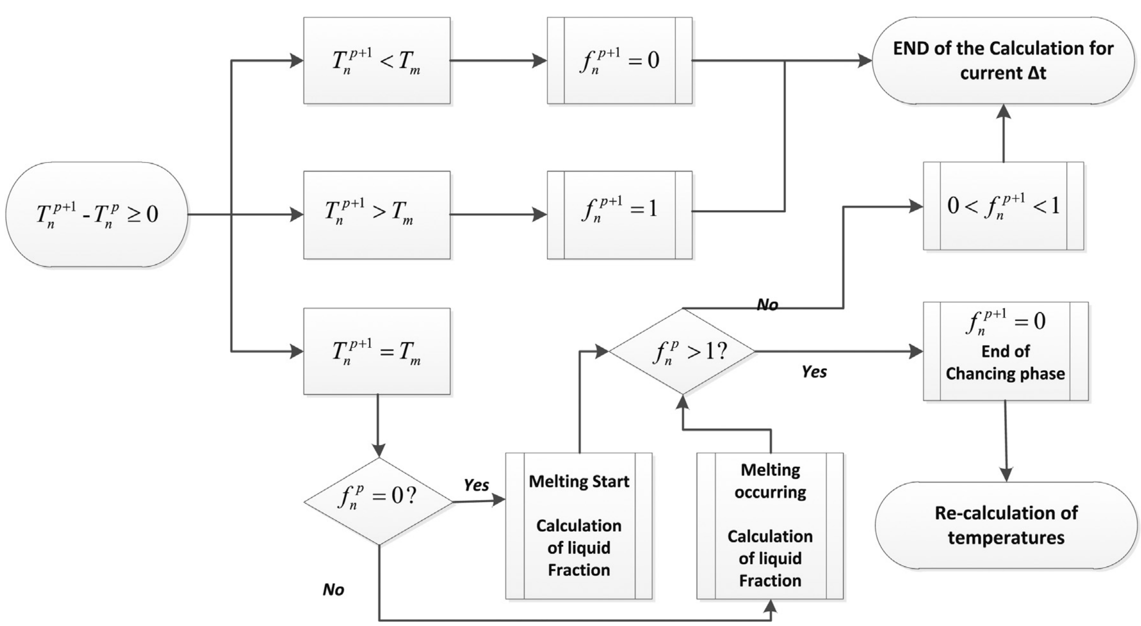

Fig. 5. Flow diagram of temperature calculation (rising temperature).

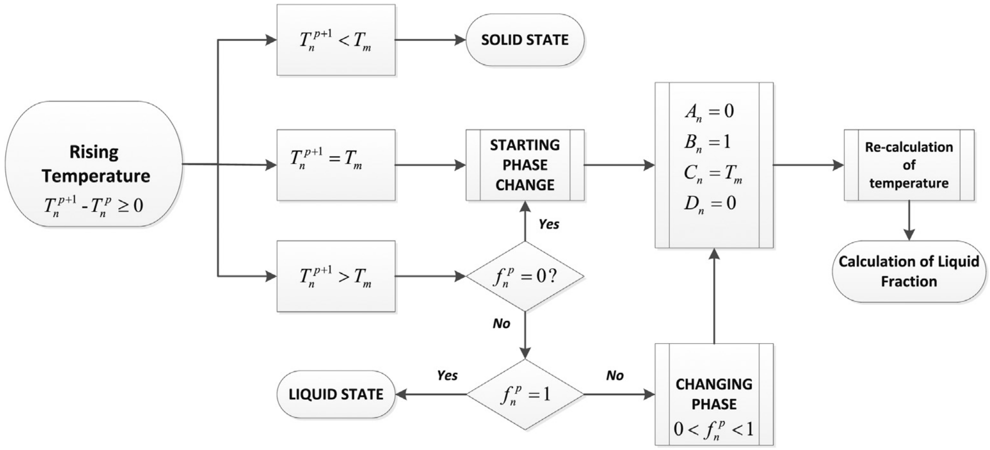

Fig. 6. Flow diagram of liquid fraction calculation (rising temperature).

storage. To evaluate the energy production of the PV panel and the resulting radiative and convective exchanges with surrounding environment, the improved five parameters model of Lo Brano et al. has been used [27] (Fig. 7).

As concerns the thermal exchanges with surrounding environment, the following assumptions were made:

- the net radiative power is:

$$q = q_{\mathrm{irr}} = \varepsilon_{\mathrm{glass}}\sigma\left(T_{\mathrm{sky}}^4 - T_{\mathrm{glass}}^4\right) + \varepsilon_{\mathrm{PET}}\sigma\left(T_{\mathrm{roof}}^4 - T_{\mathrm{PET}}^4\right)$$

where$\varepsilon_{\mathrm{glass}}$ and $\varepsilon_{\mathrm{PET}}$ are, respectively, the emissivity of the glass and PET external surfaces;σ is the Boltzmann constant;T_{sky} is the fictive sky temperature calculated with the correlation of Swinbank et al. [28].T_{roof} is the temperature of the external surface of the roof. In the proposed algorithm, the value of T_{roof} to a first approximation is T_{air}, the external air temperature.

- the heat rate "generated" (or, more appropriately, absorbed by incident radiation and not converted into electricity) in the silicon layer is:

$$q = \tau\alpha I - \frac{V^2}{R}$$

where I is the solar radiation [W/m^2], V is the voltage generated by the panel, R is a pure resistive electrical load and $\tau\alpha$ is the product of

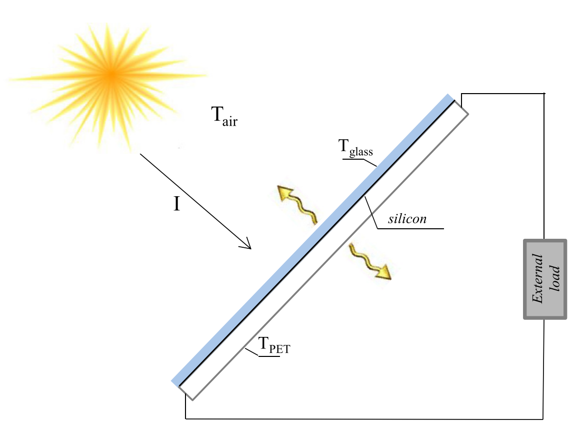

Fig. 7. Sketch of energy exchanges of PV panel with surrounding environment.

transmission coefficient of glazing and absorption coefficient of silicon.

The algorithm presented in the previous section was easily implemented in a software procedure that uses a common programming language like VB.NET; the equations presented in this section were used as concerns the simulated physical scheme.

5. Description of experimental system

To perform the comparison between measured and calculated data, an experimental system was built up and situated on the top of the Energy Department of University of Palermo, (38°07′ N, 13°22′ E). The test facility consists of a silicon PV panel, a precision resistance set, a first class (ISO 9060) Delta Ohm pyrometer mod. LP PYRA 02 AV linked to a data acquisition module Advantech ADAM 6024. A Davis Vantage PRO2 Plus Weather station was used to collect the measurements of air temperature, relative humidity, wind speed and direction, horizontal global solar irradiance and atmospheric pressure.

A Kyocera KC175GHT-2 PV panel was coupled with a PCM layer in the bottom part using a perforated metal mesh, bolted into the frame of the panel. A galvanized hexagonal iron wire mesh with a distance of mesh holes of 54 mm and a wire thickness equal to 0.9 mm was used.

The PCM was encapsulated by a double package of plastic bag, as shown in Fig. 8. The thickness of the two envelopes is equal to 0.4 mm.

The silicon temperature was measured using thermocouples (type T, copper-constantan [1]) installed into little holes made in the PET rear film of the panel. All data were collected every 30 min and stored for further calculations and ex-post processing.

To measure the electrical power produced by the PV panel, the electrical circuit has been closed to precision resistances (Vishay RH250) with a tolerance of ±1% and a temperature coefficient of ±50 ppm/°C. Since the resistances never exceeded a temperature of 150 °C, their nominal values were considered known within the precision of ±1.625% [27].

6. Results

To verify the reliability of the proposed calculation algorithm, the PV–PCM system above described was monitored during the summer season, when the system is subjected to the higher solar irradiance values. Several numerical simulations were performed

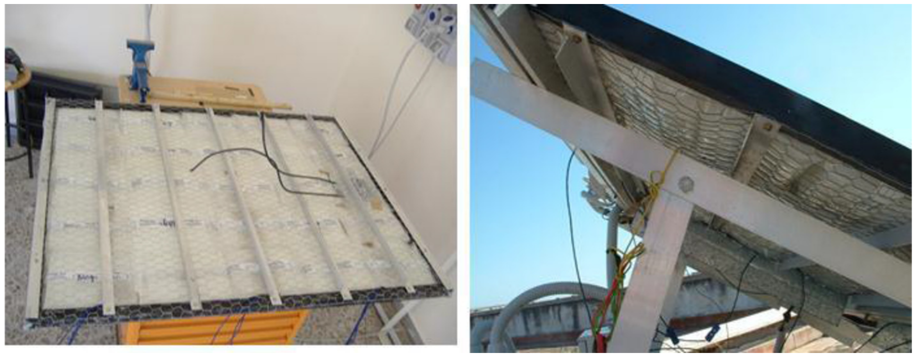

Fig. 8. Some photos of the experimental system.

Table 1
Thermo-physical characteristics of the PCM.

PCM		
Transition phase	26–28	°C
Solid density	0.87	kg/l
Liquid density	0.75	kg/l
Heat capacity	179	kJ/kg
Specific enthalpy of phase change	1.8–2.4	kJ/kgK
Thermal conductivity	0.2	W/mK

on different days and the results were compared with measured data in terms of silicon layer temperature.

As concerns the PCM, Rubitherm no. RT-27 packages were adopted, whose chemical composition was alkane hydrocarbons with the generic formula C_nH_{2n+2}. Melting point of pure paraffin depends on the number of carbon atoms, this number is between 14 and 40 and melting temperature range is between 6 °C and 80 °C. In the experimental application that was accomplished by the authors, the tested paraffin was characterised by a relatively low phase transition starting temperature (26 °C), that could be interesting for the typical Sicilian summer and autumn thermal regimes.

In Table 1, the thermo-physical characteristics of the adopted PCM are listed.

Figs. 9, 11, 13 and 15 represent the climatic conditions registered in four typical short summer season periods in Palermo, some of them are cloudy days while other are sunny days.

Figs. 10, 12, 14 and 16 show the comparison between the measured temperatures and those calculated by the proposed algorithm.

The high temperatures measured for the PV cell and shown in Figs. 10, 12, 14 and 16 testify a poor performance of the paraffin as a heat storage medium. Due to the low capability of the PCM to discharge the surplus heat during the night, (as consequence of its low thermal diffusivity) the paraffin reveals efficient in cooling the panel only for few days after the installation; this result, although controversial in literature [9,29,30] has been verified experimentally in previous works.

It may be observed that a flat power profile occurred during the hours with highest irradiance; this is due to the reaching of

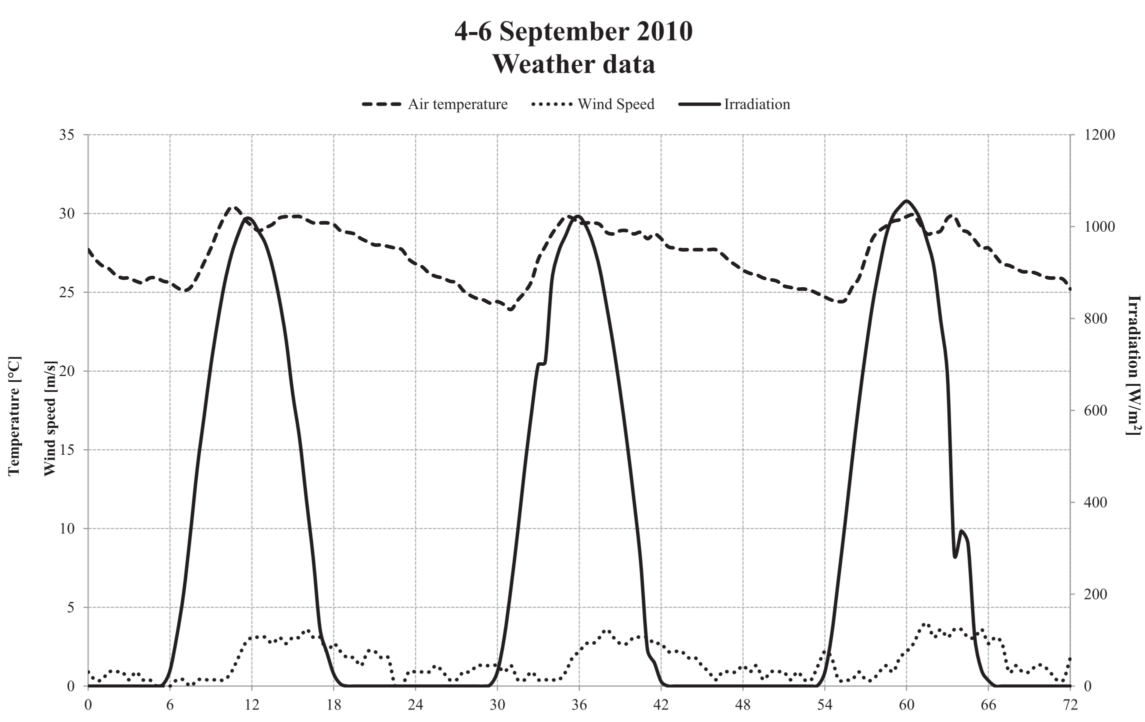

Fig. 9. Trends of climatic parameters in Palermo, September 4 to 6.

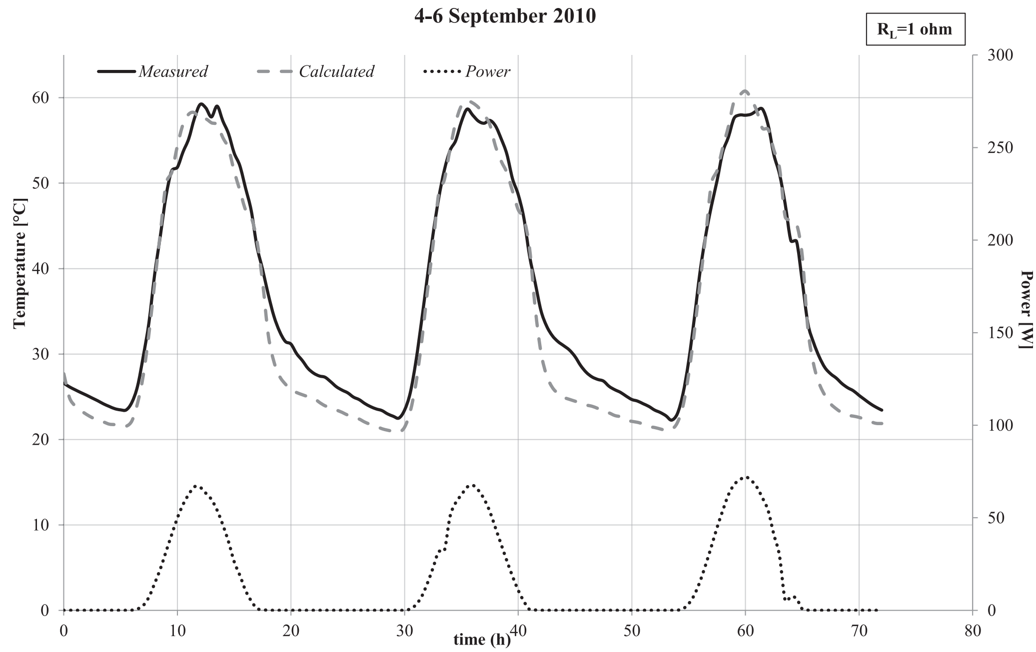

Fig. 10. Temperature trend of the measured and simulated PV–PCM system together with measured power output, in Palermo, September4 to 6.

saturation conditions of the PV panel (for 5, 10 and 18 Ω of pure resistance), where significant changes in solar irradiance do not induce variations in the power output. The attention is focused on these particular operating conditions since, under saturation conditions, the temperature of the PV panel experiences wider fluctuations being inhibited any further increase in power output delivery.

The finite differences calculation model of the PV panel coupled with a PCM thermal storage system has achieved excellent results in all cases, for sunny and partially cloudy days. The average percentage gap is calculated as $T_{calculated} - T_{measured}/T_{measured}$, while the absolute value is calculated as $|T_{calculated} - T_{measured}|/T_{measured}$.

Looking at the graphs represented before, it is possible to see as the calculated temperature trend's is in good agreement with measured temperatures, validating the reliability of the calculation model.

The calculated temperatures at night are significantly lower than the comparable measured: this is an indication of an incorrect estimate irradiative heat exchange, which were estimated under the assumption of the sky always clear. Table 2 shows that the

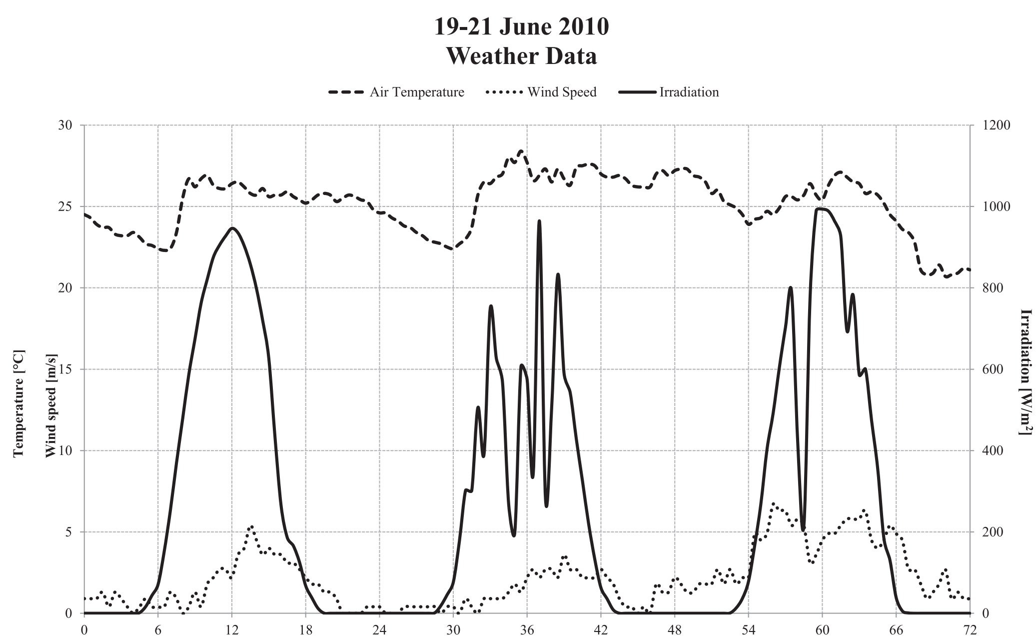

Fig. 11. Trends of climatic parameters in Palermo, June 19 to 21.

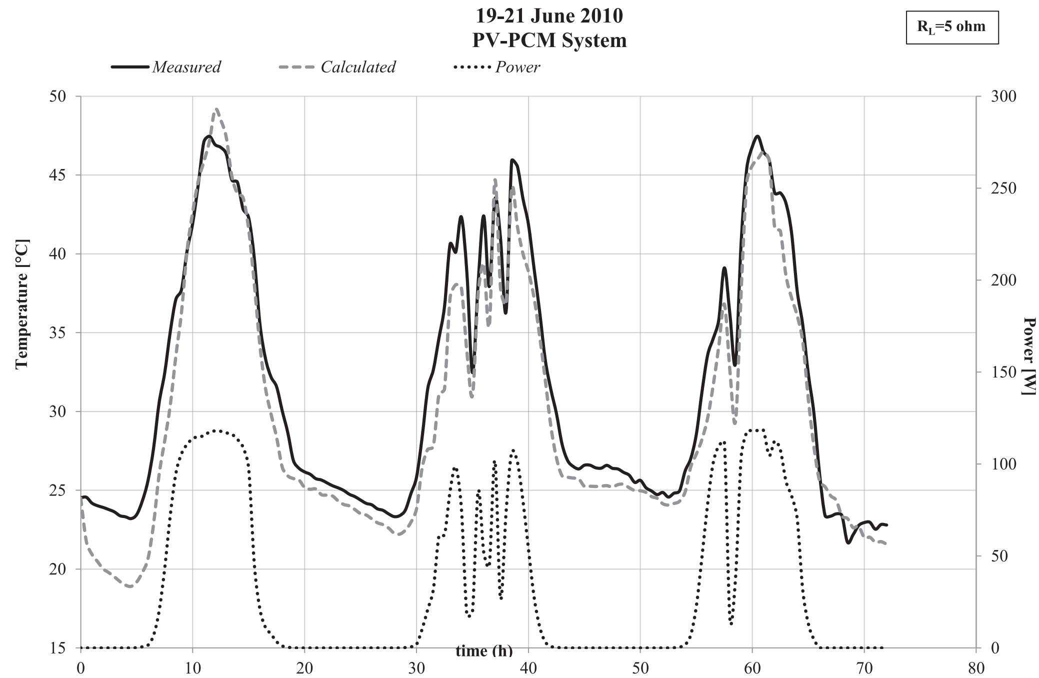

Fig. 12. Temperature trend of the measured and simulated PV–PCM system together with measured power output, in Palermo, June 19 to 21.

differences between the measured and calculated values are acceptable.

7. Conclusions

The heat exchange between the PV–PCM system and the surrounding environment is governed by several variables such as: the thermo-physical properties of all the material composing the system, the geometry, the weather conditions, the heat transfer coefficients (radiative and convective). The aim of this paper was to develop a simplified numerical model, relying upon the assumption of a one-dimensional geometry and was proposed for a PV system coupled with a PCM-based heat storage. The model is based on two sets of recursive equations that apply to two distinct type of spatial domains: internal domains with a length equal to Δx and a representative node located in the middle, and a boundary domain with

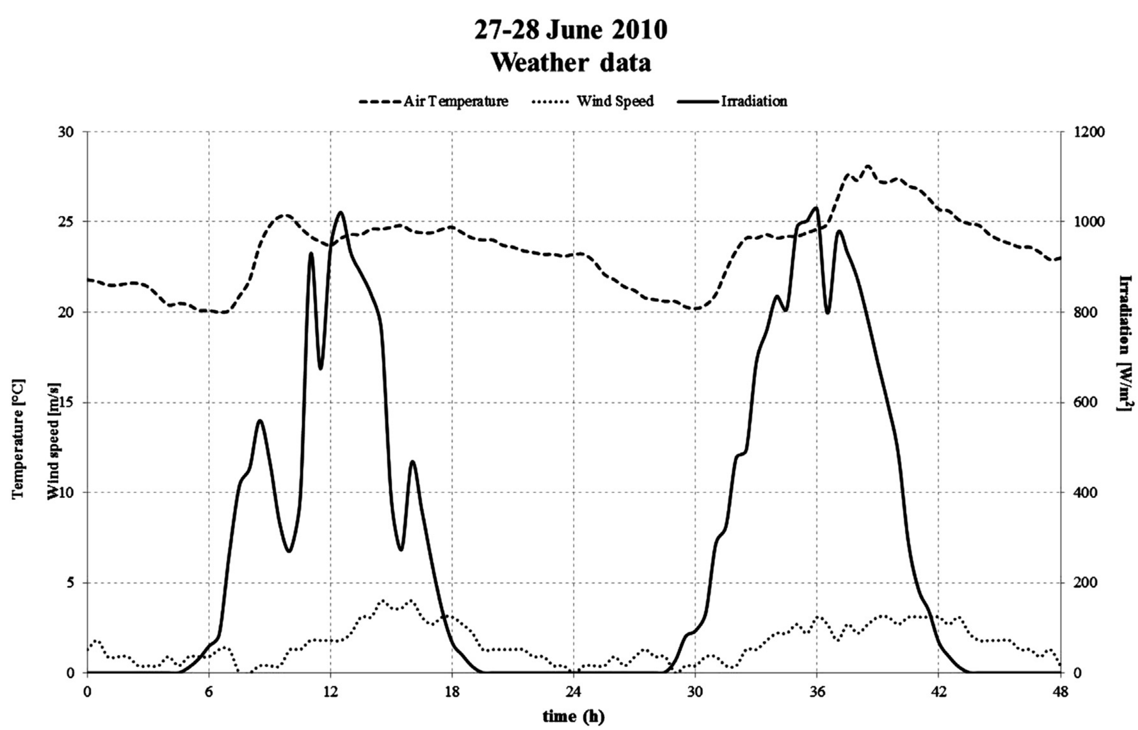

Fig. 13. Trends of climatic parameters in Palermo, June 27 to 28.

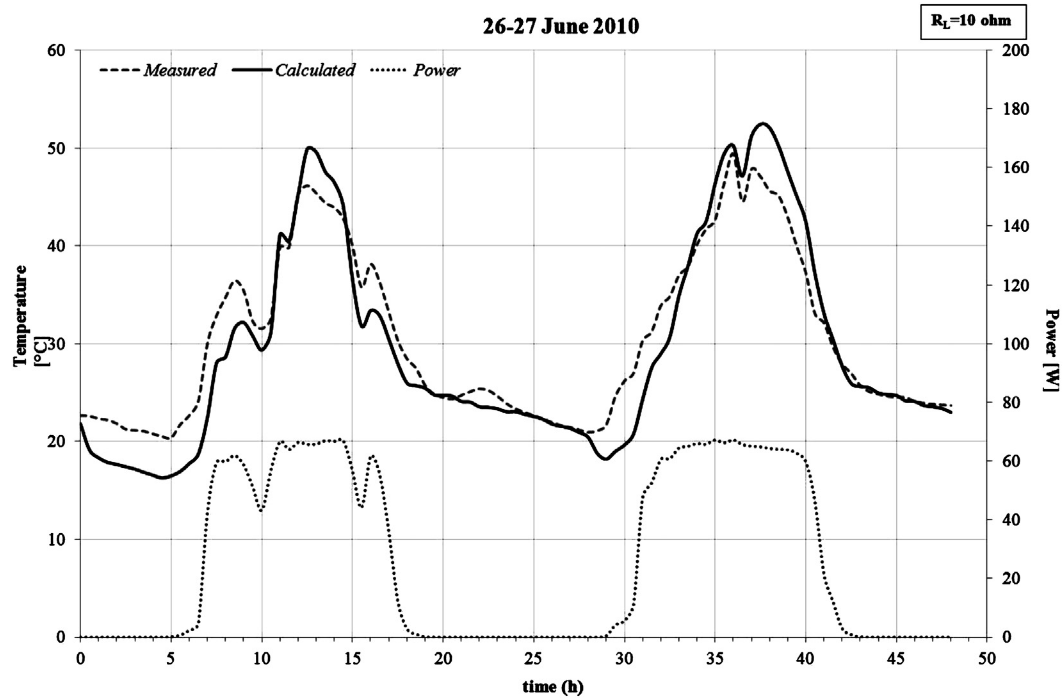

Fig. 14. Temperature trend of the measured and simulated PV–PCM system together with measured power output, in Palermo, June 19 to 21.

halved length ($\Delta x/2$) and a representative node placed on the surface. The method assumes the phase change to be strictly isothermal and updates, at each time step, the liquid fraction of domains and the temperature of corresponding nodes. A comparison was performed between the numerical results achieved by the proposed algorithm and the experimental data obtained in situ on a test facility. Despite the adoption of a simplified approach, relying upon the assumption of a one-dimensional geometry, the analysis revealed that the proposed thermal model is reliable under different climatic conditions; in the cases of sunny and partially cloudy days and different operative working conditions; in fact, the relative average gap between calculated and measured silicon temperature remained below 7%.

Data monitored in our experimental setup have demonstrated that the dominant thermal process is that one pertaining the discharge of the surplus heat during the night. A better thermal

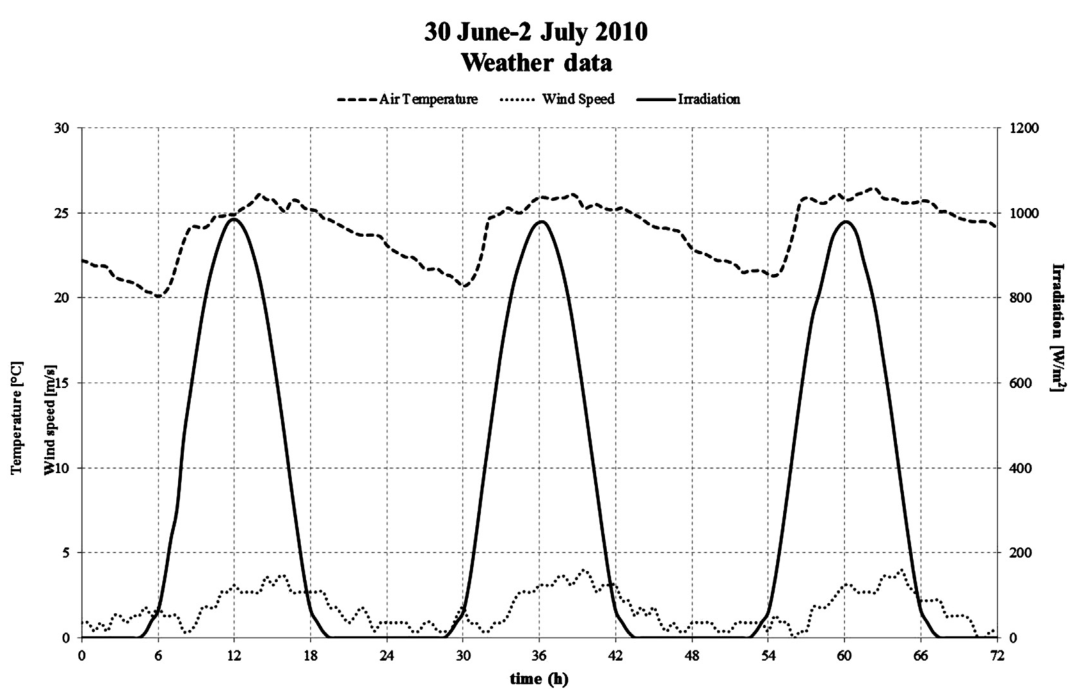

Fig. 15. Trends of weather parameters in Palermo, June 30 to July 2.

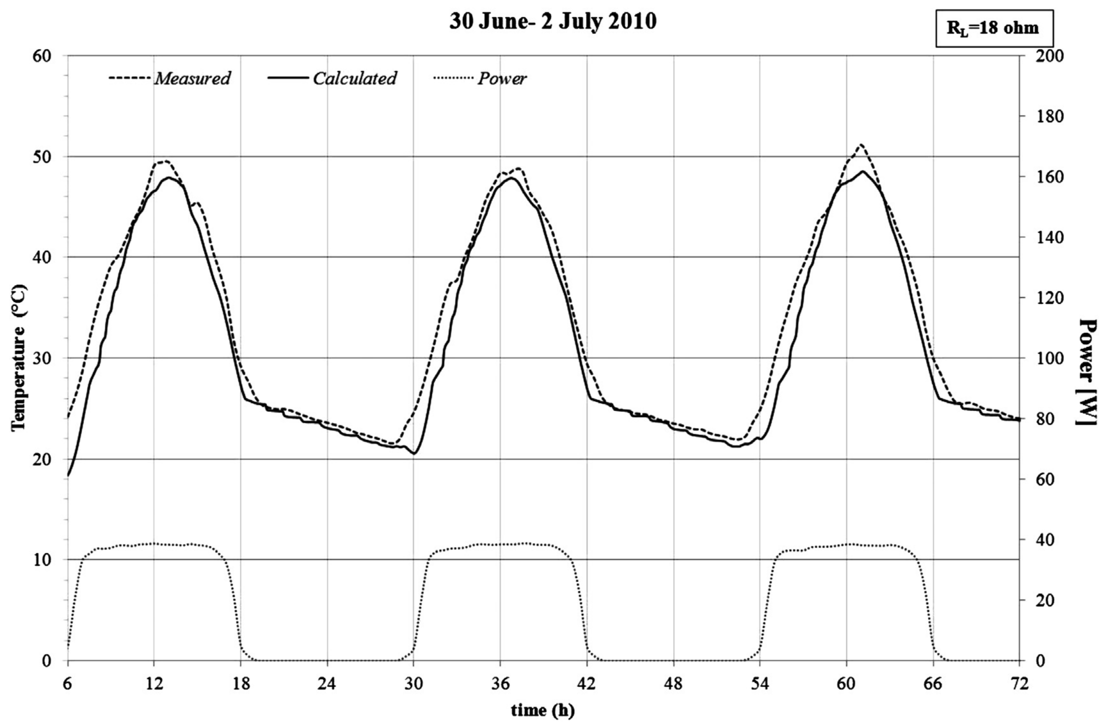

Fig. 16. Temperature trend of the measured and simulated PV–PCM system together with measured power output, in Palermo, June 30 to July 2.

Table 2
Comparison between measured and calculated operative temperature of PV–PCM system.

Operative temperature	September 4th–6th	June 19th–21st	June 27th–28th	June 30th–July 2nd
	1 Ω	5 Ω	10 Ω	18 Ω
Relative average gap	6.07%	5.49%	4.85%	4.90%
Absolute relative average gap	7.36 %	6.11%	8.37%	4.92%
Maximum negative gap	−3.24 °C	−2.21 °C	−6.43 °C	−0.11 °C
Maximum positive gap	6.07 °C	5.22 °C	7.55 °C	6.01 °C

contact would improve the heat transfer between the rear surface of the PV and the heat storage device. However, the main limit of this system is related to the heat transfer between the PCM heat storage and the surrounding environment.

To improve the accuracy of the simulation, the authors are working on a more sophisticated calculation scheme using the Crank–Nicolson approach [31]. Although this method is more complex to implement in software, this approach should allow us to obtain a good simulation using less computation time. Finally, the presented method can be used even for other PCM configurations such as those one employed in civil structures to improve the thermal performance of buildings envelope.

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